

Business Statistics

BBA — 3rd Semester — 2024 — Answer Sheet

Subject Code: STT201

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

Brief Answer Questions: $[10 \times 1 = 10]$

1. Find the quartile deviation if first and third quartiles are 18 and 30 respectively.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Calculate the combined mean from the following information.
- | Group A | Group B | Mean | Number of observations |
|---------|---------|------|------------------------|
| 6 | 5 | 6 | 5 |
| 6 | 5 | 6 | 0 |
| 6 | 5 | 6 | 0 |

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. The coefficient of correlation between two variable X and Y is 0.48. The covariance is 36. The variance of X is 16. Find the standard deviation of Y.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. In a moderately asymmetrical distribution, the mode and mean are 32 and 35 respectively. Calculate the median.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Calculate the Pearson's Coefficient of skewness when mean, mode and standard deviation are 25, 20 and 10 respectively.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Given $\lambda = 1.2$ for a Poisson distribution, find $p(x = 4)$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. If a random sample of size 36 is drawn from a finite population 400 units without replacement, then find the standard error of the sample mean if the population s.d. is 12.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Given that $P(A) = 1/3$, $P(B) = 3/5$ and $P(A \cap B) = 1/15$ then find out the value of $P(A \cup B)$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. If first quartile and third quartile of a distribution are 28 and 52 and their 90th and 10th percentiles are 61 and 17 respectively then, find the value of kurtosis.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. List out non-random sampling methods.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Short Answer Questions: (Attempt any FIVE Questions)[5 × 3 = 15]

11. The following table represents the weekly expenditure of 100 families. Expenditure 0-20 20-40 40-60 60-80 80-100 No. of Families 14 27 15 If the mean value is 48 find the missing frequencies.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. A sample of 500 bulbs of a company shows that an average life of 1400 hours with standard deviation of 30 hours. Find 95% confidence limits for population mean.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

13. Calculate coefficient of quartile deviation from the following distribution of expenditure of one thousand households of Pokhara city. Expenditure (Rs.'00) No. of households

30-50	54
50-70	100
70-90	140
90-110	300
110-130	230
130-150	125
Above 150	51

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

14. The mark distribution of 104 students are given below. Marks0 - 1010 - 2020 - 3030 - 4040 - 5050 - 6060 - 70No. of Students7813293593Find the lowest marks of the top 10% of the students.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

15. Five dice were thrown for 96 times. The observed number of times for getting 4, 5 or 6 was recorded by the table below. Fit the binomial distribution for the given data.

No. of face	4	5	6
Observed frequency	21	10	26
	3	5	15
	8	1	5

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

16. A committee of 5 is to be formed out of a group of 8 boys and 7 girls. Find the probability that in the committee.
a. There will be all boys.
b. There will be all girls.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section C

Long Answer Questions: (Attempt any THREE Questions)[3 × 5 = 15]

17. Calculate the percentile coefficient of kurtosis from the income distribution of sixty-two randomly selected employees of Nepal Bank Limited. Also interpret the result.
Income (Rs '000) 20-30 30-40 40-50 50-60 60-70
Total No. of employees 710 201 876 2

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

18. The following table shows average marks of students in two classes.
Class A **Class B**
No. of students 150 100
Average marks 45 43
Standard deviation of marks 54
Test whether there is any significant difference in average marks of students of two classes at 5% level of significance.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

19. The running capacity of two horses is given below, state which is more consistent and why?
Horse A 250 255 280 290 295 300
Horse B 280 282 290 295 298 295

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

20. Students of a class are given a test. Their marks were normally distributed with mean 60 and standard deviation 5. What percentage of students scored:
a. More than 65 marks.
b. Between 50 and 65 marks.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section D

Comprehensive Answer Questions:[20]

21. The following table gives the marks in mathematics and statistics of ten students.
Marks in mathematics 45 70 65 30 90 40 50 75 85 60
Marks in statistics 35 90 70 40 95 40 60 80 80 50
a. Calculate two regression coefficients.
b. Calculate two regression lines.
c. Find Karl Pearson's coefficient of correlation and test its significance.
d. Estimate the most likely marks in statistics when the marks in mathematics is 52.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.